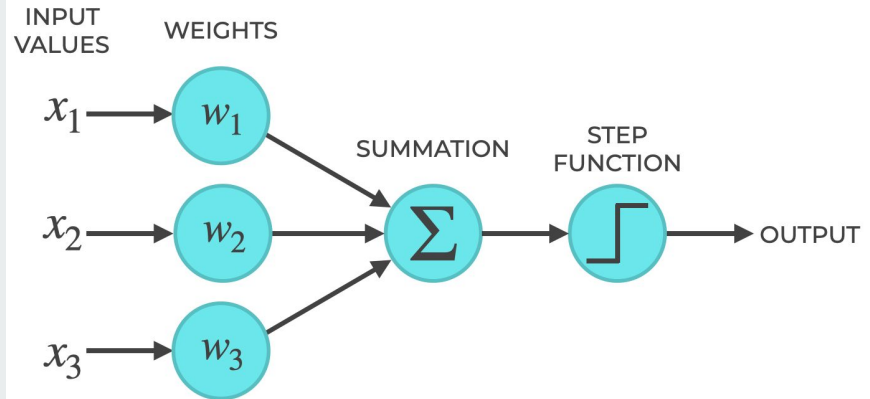


CSE 151B Discussion 1

Ben Bao
6/29/2026

THE STRUCTURE OF A PERCEPTRON





Roll Call

- Not for a grade!



What Discussions Will Cover

- Demos
- Assignment Overviews
- Reminders
- Tips & Tricks
- QnA Sessions
- Every Monday and Wednesday 2PM - 2:50PM COA B30



Today's Discussion

- Course Setup
- Assignments Overview
- Tips for Success
- QnA
- Feedback



Piazza + PrairieLearn Setup

- Piazza Link:
<https://piazza.com/class/mq7cvtfat4t3ff/>
- PrairieLearn Link:
https://us.prairielearn.com/pl/course_instance/220261/join/5GE97TRL79

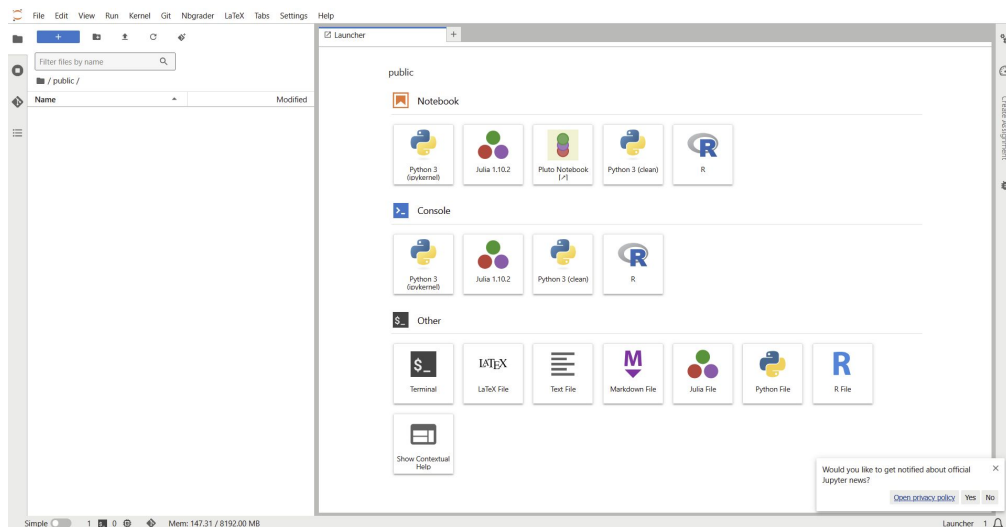


Reminder: Triton Testing Center Practice Test

- Sign up for trial run of Triton Testing Center on PrairieLearn
 - Sign in process
 - Testing environment
 - Shortcuts and tips
- **Required before you are able to sign up for any other exams in this class**
- Due by end of this week

Datahub Setup Demo

- datahub.ucsd.edu
- Main development platform for this class
- Cloud based
- CPU, RAM, and GPU compute resources





Reminder: Pre-Course Survey

- Helps us gauge student experiences and needs
- Please fill out ASAP!
- Note 9 on Piazza or at
<https://forms.gle/i4iWa6ARciUnNP9B8>



Ao Overview

- Set of 5 questions reviewing decision boundaries and derivative rules.
- Multiple choice on PrairieLearn
 - Register on Piazza @note 10
- Due 7/1 (this **Wednesday!**)

PrairieLearn CSE 151B, SS1_26 Assessments Gradebook A1

A1.1. Logistic Regression

A logistic regression model predicts $\hat{y} = \sigma(\mathbf{w} \cdot \mathbf{x} + b)$ and assigns class 1 when $\hat{y} \geq 0.5$. A student argues that because σ is a nonlinear function, the resulting decision boundary between the two classes must be nonlinear. What is your assessment of this argument?

- ☐ (a) The student is correct whenever the input features have not been standardized.
- ☐ (b) The student is correct: the steepness of the sigmoid controls the curvature of the boundary.
- ☐ (c) The student is wrong: the decision boundary is where $\mathbf{w} \cdot \mathbf{x} + b = 0$, which is a hyperplane (linear in $\mathbf{x}$).
- ☐ (d) The student is wrong, but only because the bias term b cancels the sigmoid's nonlinearity.
- ☐ (e) The student is correct: the sigmoid bends the boundary into a curved surface in $\mathbf{x}$ -space.

Save & Grade Single attempt

Save only

Additional attempts available with new variants •



Ao Example Problem

(c) $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ where $f(x, y) = e^{xy} + y \log 3x$

(c) $y(e^{xy} + 1/x), xe^{xy} + \log 3x$



Oral Exams

- On **Thursday** - sign up on Google Calendar by **TOMORROW NIGHT**
- Will have one “understand” question
- Will have one “experiment” question



Learning Process and Resource Reflection (LPRR)

- Due on **Friday**
- 6 questions, ~300 word response
- More details can be found on the syllabus under “Learning Process and Resource Reflection”
- Asks you to describe your process & tools for completing the programming portions of your assignments.



Pre-Lecture Homework (Correctness)

- Same as pre-lecture homework but now graded for correctness
- Due **Friday**



Tips for Success

- Come to class!
- Do the pre-lecture activities/assignments
- Come to discussion section
- Start early, start often!
- Keep pace with the material
- Talk to us - we are here to support you!



Any Questions, Comments, or Concerns?



Feedback Form

